



CompSci 230

Object Oriented Software Development

Course Information



Learning Outcomes

- ▶ You will become confident and fluent with the Java language.
- ▶ **OO programming:** describe and use the features typically offered by an object-oriented programming language, including support for classes, visibility, inheritance, interfaces, polymorphism and dynamic binding
- ▶ **OO Design:** explain and apply key design principles of object-oriented software development, including separation of concerns, abstraction, information hiding, programming to interfaces, coupling and cohesion, resilience to change, and reuse
- ▶ create simple OO design models
- ▶ **Design Patterns:** develop a small graphical user interface (GUI) based application in an object-oriented programming language; Assess and improve software design decisions based on object-oriented design objectives and principles
- ▶ **Frameworks:** describe important concepts of programming frameworks, including APIs, inversion of control, concurrency and event-driven programming



Lecturers & Tutors

▶ Lecturers

- ▶ Angela Chang (Course coordinator)
 - ▶ Email: angela@cs.auckland.ac.nz
 - ▶ Office hour: email for an appointment
 - ▶ Room: 303.414
- ▶ Paramvir Singh
 - ▶ Email: p.singh@auckland.ac.nz
 - ▶ Room: 303S.423
 - ▶ Office hours: email for an appointment

▶ Tutors

- ▶ Linlin Liang, email: llia618@aucklanduni.ac.nz
- ▶ Maxwell Zhu, email: mzhu843@aucklanduni.ac.nz
- ▶ Welathantrige Boteju, email: mbot450@aucklanduni.ac.nz

▶ Class Representatives

- ▶ If you are interested in becoming a class rep, please send an email to Angela. We will facilitate the voting for a class rep.



Assessments, Test & Exam

▶ Assessments: 30%

- ▶ Coderunner exercises 18% (1.4x12, 0.1x12 attendance)
- ▶ A1, reading, A2, and A3 – 12%

▶ Test + Exam: 70%

- ▶ Computer-based test – 20% 6pm Thursday 6 April
- ▶ Computer-based Exam – 50%

Test clash?
Let us know immediately

▶ In order to pass this course, you must achieve a pass

- ▶ (i.e. at least 50%) in the combined total of the test and exam.
- ▶ (i.e. at least 50%) of the full course total of 100%.

illness, accidents ..
for test/exam
Contact Student Health and
Counselling to apply for an aegrotat



Schedule

Week	Tue	Wed	Thur	
1: 27 Feb	Lecture 1	Lecture 2	Lecture 3	Lab01 (1.5%)
2: 6 Mar	Lecture4	Lecture5	Lecture 6	Lab02 (1.5%)
3: 13	Lecture7	Lecture8	Lecture9	Lab03 (1.5%)
4: 20	Lecture10	Lecture11	Lecture12	Lab04 (1.5%)
5: 27	Lecture13	Lecture14	Lecture15	Lab05 (1.5%), AI (2%)
6: 3 Apr	Lecture16	Lecture17	Test(20%) Thur No Lecture	Easter Holiday Lab06 (1.5%)
	Mid-semester break			
7: 24 Apr	Lecture18/19	HOLIDAY	Lecture20	Lab07 (1.5%) Reading(2%)
8: 1 May	Lecture21	Lecture22	Lecture23	Lab08 (1.5%)
9: 8	Lecture24 A2 (2% Mon)	Lecture25	Lecture26	Lab09 (1.5%)
10: 15	Lecture27	Lecture28	Lecture29	Lab10 (1.5%)
11: 22	Lecture30	Lecture31	Lecture32	Lab11 (1.5%)
12: 29	Lecture33 A3(4% Mon)	Lecture34	Lecture35	Lab12 (1.5%)



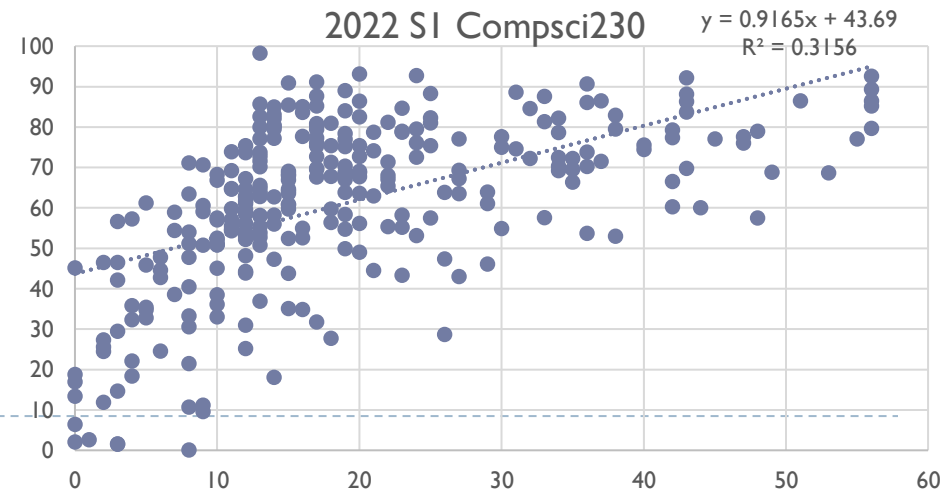
Resources

- ▶ Lecture slides, recordings, videos etc
 - ▶ Note: All **marks, lecture slides, recordings, extra videos** and **announcements** can be found on the Canvas system. <https://canvas.auckland.ac.nz>
 - ▶ Read textbook material/lecture handout prior to the lecture
 - ▶ Try In-class exercise and Post-lecture Quizzes after class (Part I)
- ▶ Piazza:
 - ▶ We encourage you to be an **active** participant and help your fellow students out by **answering** questions on Piazza.
 - ▶ **Full lines of code** should **never** be posted on Piazza, and hints rather than solutions should be given.
 - ▶ Avoid duplicated questions: **Search existing posts** before adding a new one.
- ▶ Recommended Reading (part I)
 - ▶ Java How to Program (late objects), 11th Edition by Deitel & Deitel, Prentice Hall
 - ▶ Not compulsory but comes strongly recommended.
 - ▶ Additional Resources
 - How to set up Java, Notepad++, Textpad, ...etc
 - ▶ Required readings are online, mostly from **The Java Tutorials**
 - ▶ Start today! Read about Object-Oriented Programming Concepts, at <http://docs.oracle.com/javase/tutorial/java/concepts/index.html>
 - ▶ If you know Python, read **Java for Python Programmers** <http://fpl.cs.depaul.edu/jriely/java4python/>

How to do well ...



- ▶ Be proactive, creative, learn teamwork
- ▶ Study lecture notes before AND after lecture
- ▶ Read the recommended online textbook
- ▶ **Prepare for the labs (read and think about questions)!**
- ▶ Do the labs/assignments yourself, do optional CodeRunner quizzes/exercises
- ▶ Tips to Learn Programming
 - ▶ Grasp the fundamentals for long-term benefits
 - ▶ Code by hand: It sharpens proficiency
 - ▶ Ask for help (Piazza, tutors/demonstrators, lecturers)
 - ▶ Seek out more online resources, create your own tasks.
 - ▶ Don't just read the sample code. Tinker with it!
- ▶ Regular note taking and journal writing





Labs/Tutorials/Code Runner Exercises

- ▶ Lab02 – Lab12 (in campus)
 - ▶ Lab01 Revision DIY
- ▶ What happens at a lab:
 - ▶ Download lab worksheet from Canvas, discuss and answer them briefly
 - ▶ Work on your CodeRunner questions and submit online before the due date
 - ▶ Submissions are graded by running a series of test cases of the code in a sandbox, comparing the output with the expected output.
 - ☐ You will see immediate feedback and your score once you run your code.
 - ☐ **Develop your code** in a separate development environment (e.g., **Notepad++**) and test it there until you feel you have gotten all the bugs out.

Expected	Got	
Hello, world	Hello	✗

Your code must pass all tests to earn any marks. Try again.

Incorrect

Marks for this submission: 0.00/1.00.

Expected	Got	
Hello, world	Hello, world	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.



CodeRunner: Extra Features

► Grade Report

- Shows all graded activities where your grades have been released, as well as feedback from your instructor (if any).

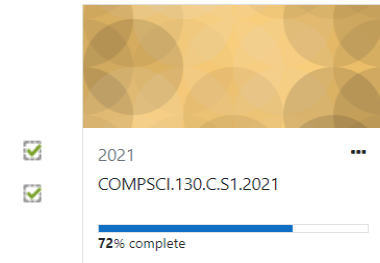
► Checking your progress:

- Activity Completion allows you to check your progress.
- When you complete a quiz and earn a score of at least 75%, the quiz will be marked with a green tick.

Labs (20%)

 [Lab01 - Introductory Lab: Python fundamentals \(Re-open for revision purpose\)](#)

 [Lab 02 - Modularity \(Re-open for revision purpose\)](#)



► Earning Badges

- When you complete a group of quizzes and earn a score of at least 75% each, you will be awarded with a badge.

	CS130L01	lecture01 exercises	Users are awarded this badge when they complete the following requirement: • The following activity has to be completed: ◦ "Quiz - Lecture01 - Python Extra Topics"
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Panopto Video

▶ Watching:



- ▶ play/pause a video, scan through using the progress bar, and easily rewind by 10 seconds if you miss something.
- ▶ Adjust the play speed if you need to.

▶ Captions:

- ▶ Clicking the “CC” button will turn the captions on/off. It is a toggle switch and will turn captions on if they are off, and off if they are on.

▶ Thumbnails:

- ▶ Navigate via the slide thumbnails at the bottom of the screen.

▶ Contents:

- ▶ See the table of contents entries, which include timestamps.
- ▶ Select an entry to jump to the timestamp in the video.

▶ Rating:

- ▶ Rate videos by using the five star button at the top menu.



Personalising a video

► Search Bar:

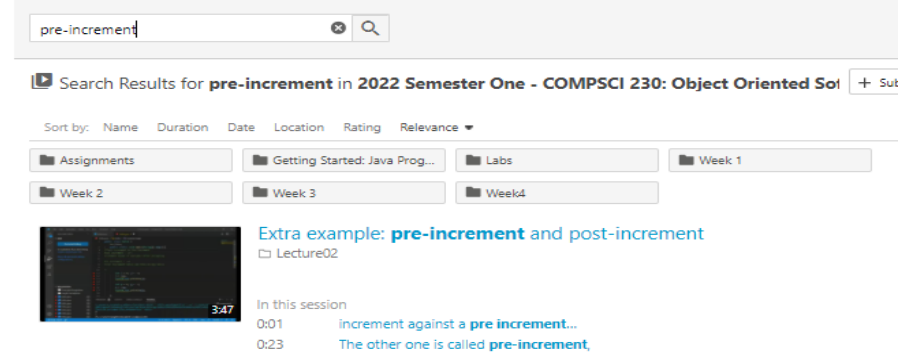
- Search a specific topic, tag, or any word mentioned in the presentation or transcript from all our videos.
 - Jump to the specific section of the video

► Bookmarks:

- Add bookmarks to a video to highlight key points.

► Notes:

- Make notes on a video directly in Panopto,.
 - Keep these notes private, only visible to you, or you can collaborate on notes with your peers and use shared channels to keep notes.





Kia atawhai! - Be kind!

Discussion Forum (Piazza) Guidelines

- ▶ Please be respectful of your peers and others in your posts.
 - ▶ Any offensive language and you will be banned.
- ▶ We encourage you to be an active participant and help your fellow students out by answering questions on Piazza.
 - ▶ **Full lines of code should never be posted** on Piazza, and hints rather than solutions should be given.
- ▶ Questions about homework wording, lecture slides, or class logistics are welcome and encouraged;
- ▶ Avoid duplicated questions:
 - ▶ **Search** existing posts before adding a new one.



Policy on Cheating and Plagiarism

- ▶ We use many ways to check that the work each student submits for marking is **their own work** and was not produced by, or copied from, someone else.
 - ▶ We start our checks by **running** a comparison program, which automatically compares all submissions from students.
- ▶ Note:
 - ▶ All assignments deemed to be too similar will be assigned a **zero mark**, and will be invited (by email) to discuss the situation with the course supervisor.
 - ▶ Offenders may be referred to the University Disciplinary Committee. See <http://www.auckland.ac.nz/ua/home/about/teaching-learning/academic-integrity>.
- ▶ Both the person who copied the work **and the person whose work was copied** are allocated a zero mark.
 - ▶ It is important that you **do not lend your assignments to others**. Never give anyone a copy of your assignment. It is the responsibility of each student to ensure that others do not copy their work.



Expectations

- ▶ COMPSCI 230 is a 15 point course
- ▶ The Faculty of Science expects an average student who is expecting an *average grade* to spend 150 hours on the course.
- ▶ There is statistically significant correlation between lecture attendance and final grade
- ▶ Expectations
 - ▶ We expect attendance at every lecture.
 - ▶ We expect attendance at every lab.
 - ▶ We expect at least 5 hours per week of practice outside class
 - ▶ We expect you to work independently and find out information you don't know
 - ▶ Read Student Expectations Document (download it from Modules)